

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination

December 2013

CL101.01 Fundamentals of Civil Engineering

Date: 02.12.2013, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Explain water conveyance system. [07]

Q - 2 (a) Explain two fundamental principles of surveying with examples. [06]

(b) Attempt any Two. [08]

(i) An area of 49 cm^2 of a map represents an area of 2401 km^2 . Find scale and RF of a map.

(ii) Convert following WCB into Reduced Bearings: 49° , 240° , 133° , 335°

(iii) Describe indirect or reciprocal ranging.

Q - 3 (a) A 20m chain was found to be 8cm too long at the end of day's work after measuring 4000m. If the chain was correct before the commencement of the work, find the correct length of the line. [04]

(b) Attempt any Two. [10]

(i) The following offsets were taken at 10m intervals from a survey line to an irregular boundary line: 2.50, 4.40, 6.60, 5.50, 7.40, 8.70, 7.80, 6.50, 4.30, 3.20 m

Calculate the area enclosed by survey line, the irregular boundary line and the first and last offsets by i) The trapezoidal rule ii) Simpson's rule

(ii) The following bearings were taken of a closed traverse ABCD. Calculate interior angles of the traverse.

Line	FB	BB
AB	$45^\circ 00'$	$225^\circ 00'$
BC	$123^\circ 30'$	$303^\circ 30'$
CD	$181^\circ 00'$	$1^\circ 00'$
DA	$289^\circ 00'$	$109^\circ 00'$

(iii) The following consecutive readings were taken with a dumpy level along a chain line

at a common interval of 5m. The position of the instrument was changed after 3rd, 7th and 10th readings. Draw out the form of a level book and enter the above readings properly. Assume the RL of the 1st point as 50.000. Calculate RL of all points and apply usual checks.

SECTION - II

- Q - 4 (a) Enlist minimum 8 branches of civil engineering. [04]
 (b) Write in brief about various loads acting on buildings. [03]
- Q - 5 (a) Discuss basic requirements of building planning. [07]
 (b) Attempt any One [07]
 (i) Write classification of building based upon occupancy and structure.
 (ii) Sketch lay out of an industrial building showing maximum details.
- Q - 6 (a) i. Write properties of cement [04]
 ii. Write uses of brick. [03]
 (b) Attempt any One [07]
 (i) Discuss suitability and requirements of different modes of transportation.
 (ii) Discuss about elements of traffic engineering.

Line	FB	HB
AB	15° 30'	325° 00'
BC	125° 30'	303° 30'
CD	181° 00'	1° 00'
DA	289° 00'	105° 00'